Cambridge (CIE) A Level Maths: Mechanics

Your notes

Kinematics Graphs

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- Drawing Travel Graphs

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Your notes

Displacement-time graphs

What is a displacement-time graph?

- **Displacement-time** graphs show the displacement of an object from a **fixed origin** as it moves in a **straight line**
- They show **displacement** (on the vertical axis) against **time** (on the horizontal axis)
- Displacement-time graphs can go **below the horizontal axis** whereas **distance-time graphs** can not
  - Distance can **not** be negative whereas displacement can be

What are the key features of a displacement-time graph?

- The **gradient** of the graph equals the velocity of the object
  - A **positive gradient** means the object is travelling **forwards**
  - A **negative gradient** means that the object is travelling **backwards**
  - The **steeper** the line, the **greater** the **speed**
- A **straight line** shows that the object is moving at a **constant velocity**
- A **curved line** shows that the object is **accelerating** or **decelerating**
- A **horizontal line** shows that the object is **stationary**
- If the graph **touches** the **x-axis,** then the object is **at the origin** at that time

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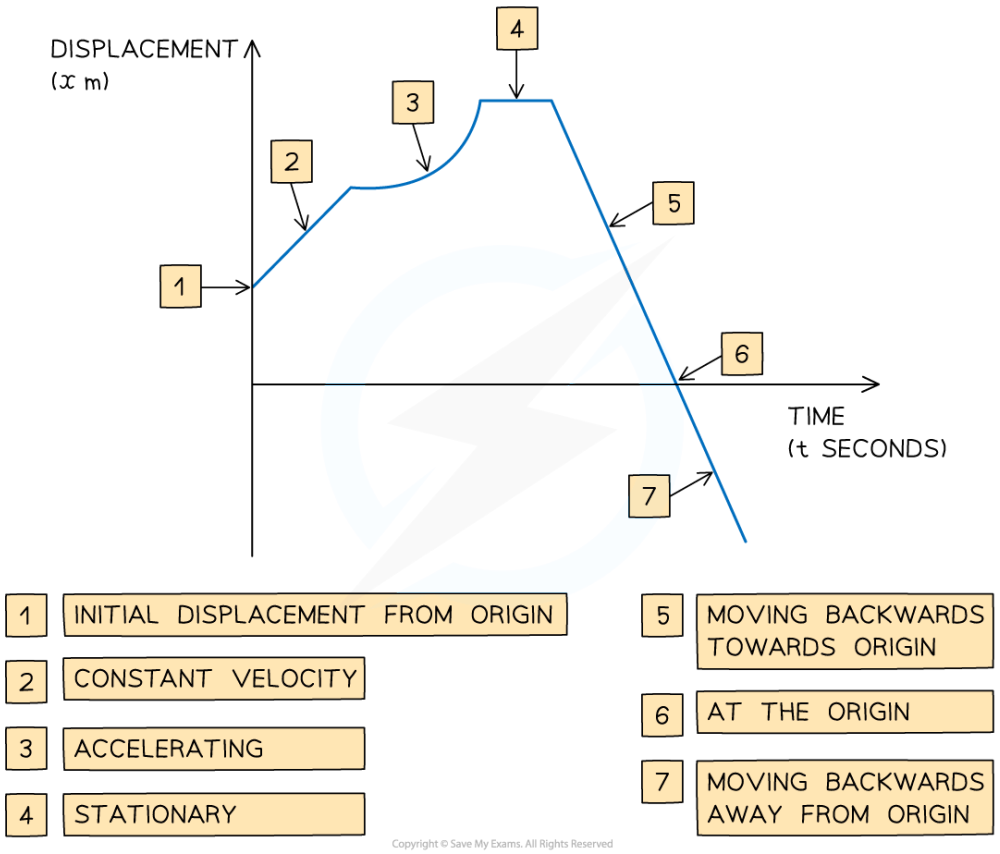

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How do I find the average speed and the average velocity of a journey?

$$\text{Average speed} = \frac{\text{Total distance travelled}}{\text{Time taken}}$$

- The average speed can **not** be **negative** as speed is a **scalar** quantity and can only take a positive value

$$\text{Average velocity} = \frac{\text{Displacement from starting point}}{\text{Time taken}}$$

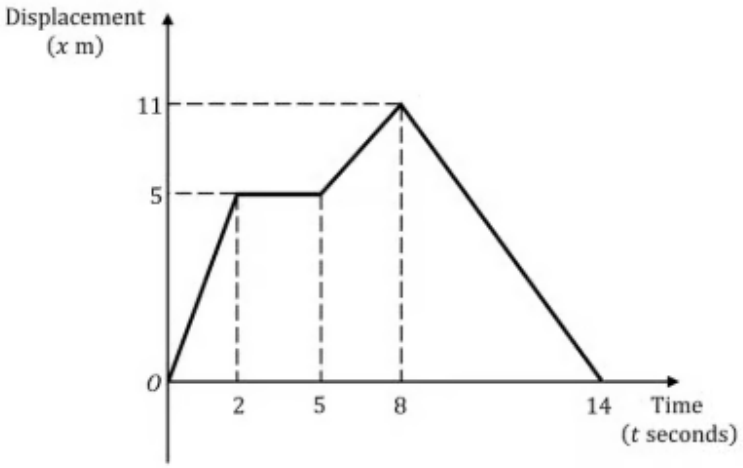
- The average velocity is a **vector** quantity and can be **positive**, **zero** or **negative**

Worked Example

An athlete is training by jogging in a straight line.

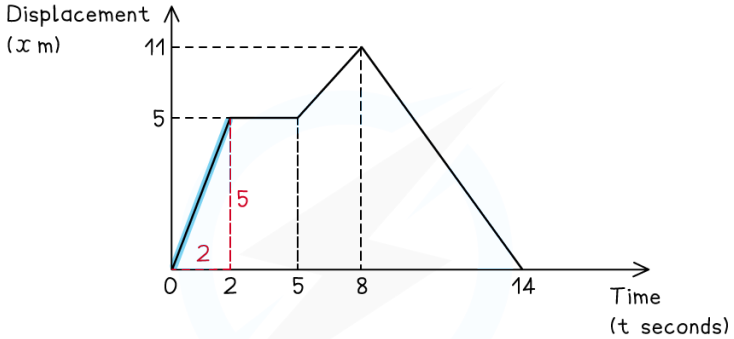
The displacement-time graph below shows the displacement of the athlete from her starting point throughout her motion.

Your notes



(a) Calculate the velocity of the athlete during the first 2 seconds.

a)



Velocity = Gradient

Velocity = $\frac{5}{2}$

Velocity = 2.5 ms^{-1}

Remember units!

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(b) Describe the motion of the athlete between the times of 2 seconds and 5 seconds.

b)

Horizontal line means stationary

The athlete is stationary

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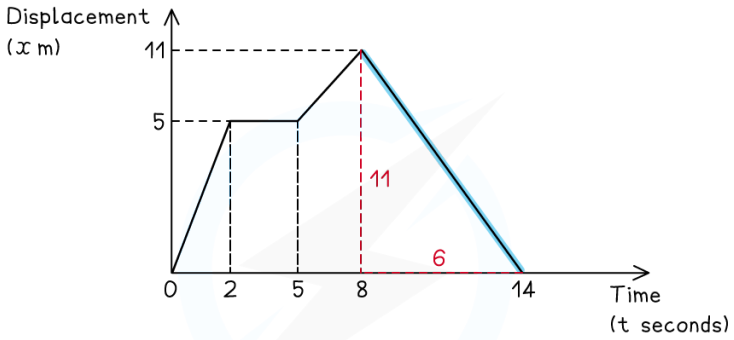
(c) Calculate the velocity of the athlete at 10 seconds.

c)



Velocity = Gradient

$$\text{Velocity} = \frac{0 - 11}{14 - 8}$$

$$\text{Velocity} = -\frac{11}{6}\ \text{ms}^{-1} = -1.83\ \text{ms}^{-1}\ (3\ \text{sf})$$

Velocity is negative as the gradient is negative

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(d) Find the total distanced travelled by the athlete during the 14 seconds.

d) Total distance = Distance forwards + distance backwards

Total distance = 11 + 11 = 22 m

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Examiner Tips and Tricks

- Be careful to spot if you are working with a **displacement-time graph** or a **velocity-time graph**.
- Be careful to spot if you are working with a displacement-time graph or a distance-time graph
- Check where the graph starts from on the y-axis, the object does not have to start at 0. For example, if the graph starts at 100, the scenario could be a student on a walk starting 100 m from their house.
- Distance is a scalar so it can **not** be negative whereas displacement from a starting point can be.
- Think about the units when calculating, make sure they are consistent.

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Velocity-time graphs

What is a velocity-time graph?

- **Velocity- time** graphs show the **velocity** of an object as it moves in a **straight line**
- They show **velocity** (on the vertical axis) against **time** (on the horizontal axis)
- Velocity-time graphs can go **below the horizontal axis** whereas **speed-time graphs** can not

What are the key features of a velocity-time graph?

- The **gradient** of the graph equals the **acceleration** of an object
- A **straight line** shows that the object is **accelerating** at a **constant rate**
- A **horizontal** line shows that the object is moving at a **constant velocity**
- The **area** between graph and the x-axis tells us the **change in displacement** of the object
  - Graph **above** the x-axis means the object is moving **forwards**
  - Graph **below** the x-axis means the object is moving **backwards**
- The **total displacement** of the object from its starting point is the sum of the **areas above** the x-axis **minus** the sum of the **areas below** the x-axis
- The **total distance travelled** by the object is the sum of **all** the **areas**
- If the graph **touches** the **x-axis** then the object is **stationary** at that time
- If the graph is **above** the **x-axis** then the object has positive velocity and is **travelling forwards**
- If the graph is **below** the **x-axis** then the object has negative velocity and is **travelling backwards**

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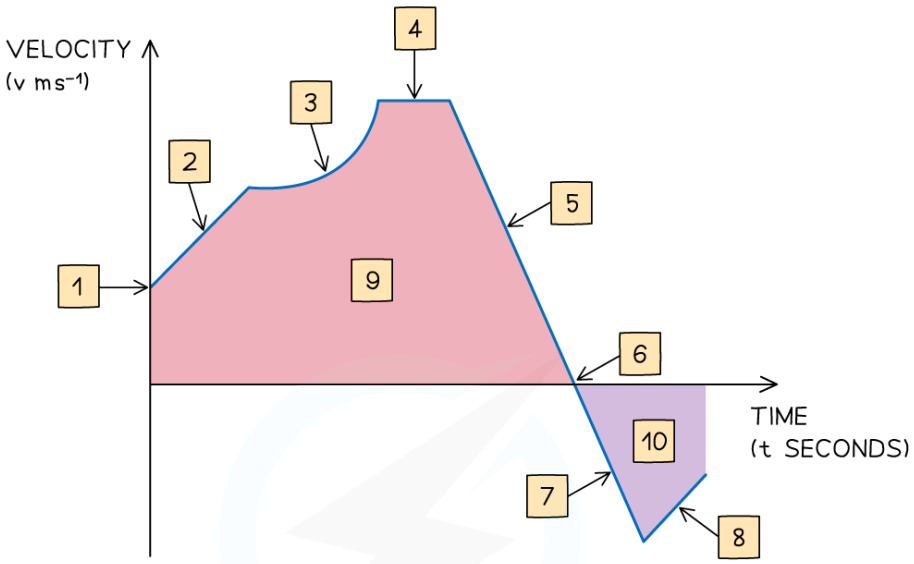

1	INITIAL VELOCITY
2	CONSTANT ACCELERATION
3	VARIABLE ACCELERATION
4	CONSTANT VELOCITY
5	DECELERATING (SLOWING DOWN BUT STILL MOVING FORWARDS)
6	INSTANTANEOUSLY AT REST (STATIONARY FOR AN INSTANT)
7	SPEEDING UP BUT MOVING BACKWARDS
8	SLOWING DOWN BUT STILL MOVING BACKWARDS
9	DISTANCE TRAVELLED FORWARDS
10	DISTANCE TRAVELLED BACKWARDS

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Worked Example

The motion of a bird travelling in a straight line is tracked for 60 seconds.

This information is displayed in the velocity-time graph below.

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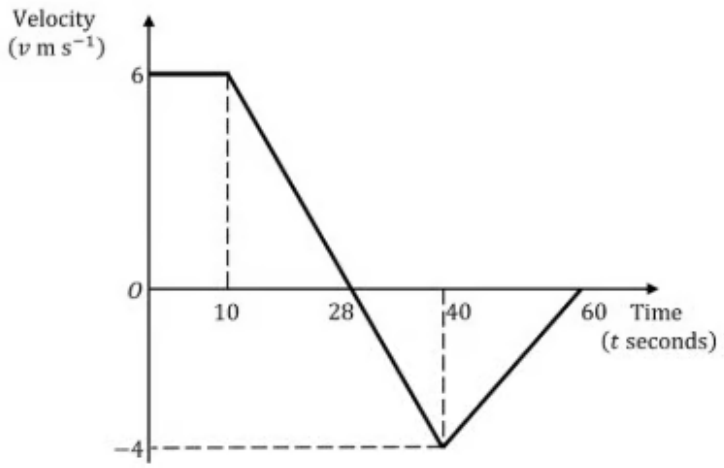
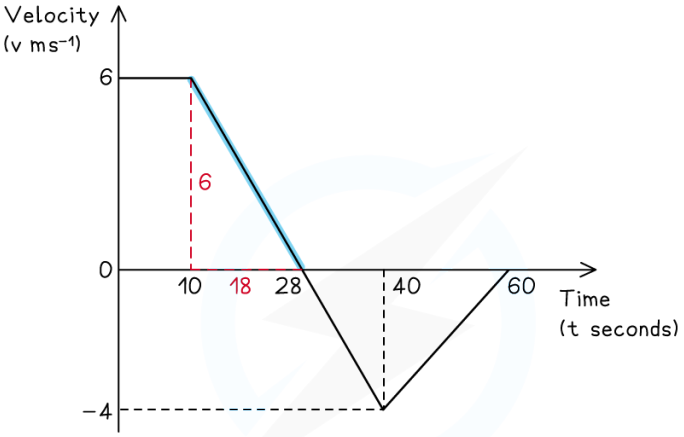


(a) Calculate the acceleration of the bird at 20 seconds.

a)



Acceleration = gradient

Acceleration = $-\frac{6}{18}$ negative as gradient is negative

Acceleration = $-\frac{1}{3}$ ms^{-2} = −0.333 ms^{-2}

Remember units!

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(b) Calculate the distance travelled in the first 28 seconds.

b) Displacement = area

Step 1: Identify the shape

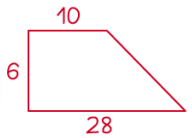
Step 2: Identify the lengths

Step 3: Find the area above x−Axis

So distance = displacement

Distance = 114m

Trapezium



$\frac{1}{2}(10 + 28)(6) = 114$

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(c) Calculate the displacement of the bird from its starting point after 60 seconds.

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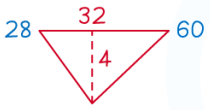
c) Distance forwards = 114 m

Step 1: Find the distance backwards by finding the area beneath the x-axis



Area = $\frac{1}{2}(32)(4) = 64$ m

THE BIRD HAS TRAVELLED BACKWARDS 64 m

Step 2: Calculate displacement by distance forwards − distance backwards
114 − 64

Displacement = 50 m from starting point

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Examiner Tips and Tricks

- Be careful to spot if you are working with a distance-time graph or a velocity-time graph.
- Be careful to spot if you are working with a speed-time graph or a velocity-time graph.
- Check where the graph starts from on the y-axis, the velocity does not have to start at 0. For example, the scenario could be a car driving at a constant speed and the driver sees a hazard.
- Be extra vigilant when working with negative gradients or with graphs under the x-axis, it is easy to make mistakes with these.
- Speed is a scalar so it can **not** be negative whereas velocity can be.

  Make sure your units are consistent.

Your notes

Drawing travel graphs

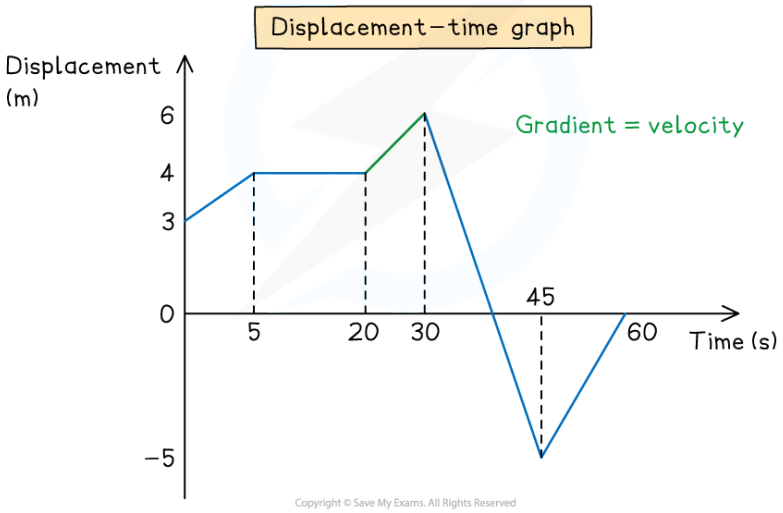
How do I draw a displacement-time graph or a velocity-time graph?

- You may be asked to draw a **displacement-time** graph or a **velocity-time** graph
- You will be given information about different sections of a journey and must draw each section and **label** the **points** on the axes
- Remember that time will always be on the x-axis
- Always remember to include **units** when you label the axis
- Use appropriate facts about the **gradient** and **area** under the graph to work backwards and find relevant or missing information
  - The **gradient** of a **displacement-time** graph is the **velocity** and the **gradient** of a **velocity-time** graph is the **acceleration**
  - The **area** between a **velocity-time** graph and the x-axis is the **displacement**

Step 1: Draw appropriate axes
(label displacement/time or velocity/time)

Step 1: Mark on key points given in the question
(label these on the axes)

Step 1: Draw in the lines to complete the graph



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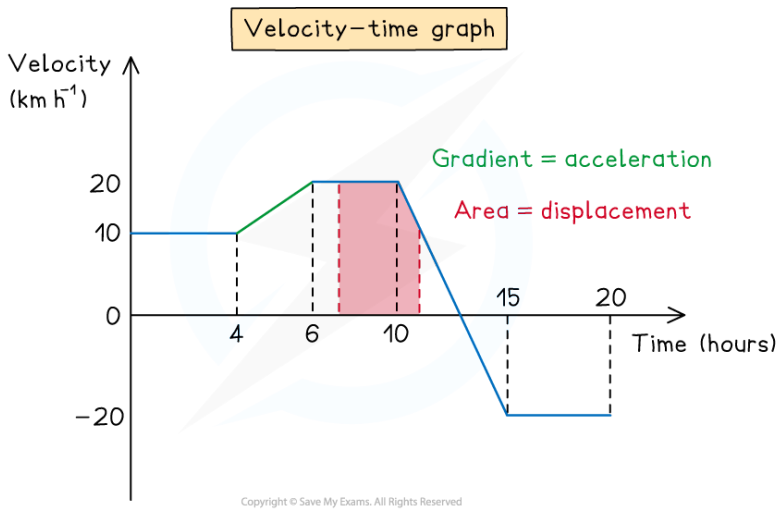
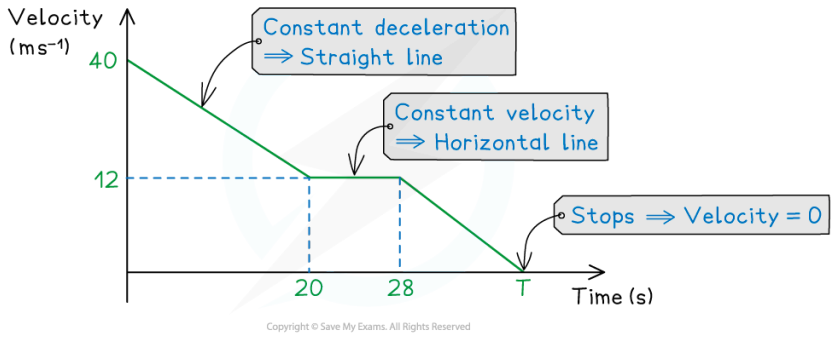


Worked Example

The driver of a train, travelling at 40 m s^{-1} on a straight horizontal track, is informed of debris on the track 0.74 km ahead and immediately applies the brakes.
The train decelerates uniformly for 20 seconds to a speed of 12 m s^{-1}
The train then maintains this speed for another 8 seconds.
The driver reapplies the brakes to slow down the train at a constant rate and bring it to a complete stop 10 m before the debris.

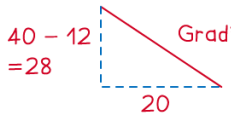
(a) Sketch a velocity-time graph to show the motion of the train.

a)



(b) Find the deceleration of the train from the moment the brakes were first applied to the moment its speed first reached 12 m s^{-1}.

b) Gradient of velocity – time graph is the acceleration



Gradient $= -\frac{28}{20} = -1.4$

Question says deceleration already so we don't need the negative sign

Deceleration = 1.4 ms^{-2}

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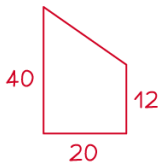
(c) Calculate the total time from the moment the brakes were first applied to the moment the train came to rest.

c) Area under velocity – time graph is the displacement

Trapezium



$\frac{1}{2}(12 + 40)(20) = 520$

Rectangle

$(8)(12) = 96$

Triangle

$\frac{1}{2}(12(T - 28)) = 6(T - 28)$

Stops 10 m before debris

0.74 km – 10 m = 730 m

Areas should add to 730

$520 + 96 + 6(T - 28) = 730$

$6(T - 28) = 114$

$T - 28 = 19$

$T = 47$

Time = 47 seconds

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Examiner Tips and Tricks

- As in the worked example, examiners can use other words, such as uniformly, to mean constant.
- Be careful to spot if you are working with a displacement-time graph or a velocity-time graph.
- Remember that displacement and velocity can be negative whereas distance and speed can **not** be negative.
- Take care when a velocity-time graph is below the x-axis, if it has a negative gradient then it is speeding up and moving backwards. If it has a positive gradient below the x-axis then it is still moving backwards but it is slowing down.
- Think about the units when calculating displacement, velocity and acceleration.

Your notes